Theory of Matrices

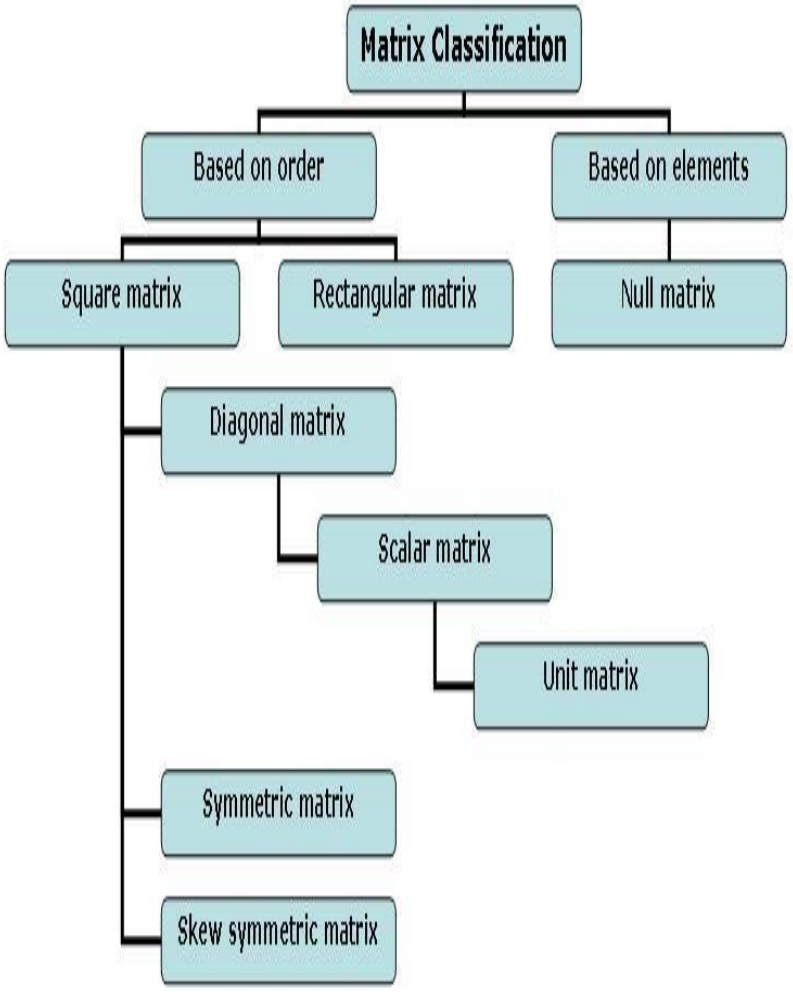
An arrangement of certain numbers in an array of m rows and n columns such as

$$\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

m – no. of rows row in A, n – number of columns in A. Thus, matrix is denoted by $A_{m\times n}$.

$A = \{a_{ij}\}_{m\times n}$ where a_{ij} denotes an element belonging to $i^{th}\ row\ and\ j^{th}\ column$.

Types of Matrices



Name	*Definition*	*Example*
Rectangular	$n \times p$	$\begin{bmatrix} 1 & 2 & 3 \\ -4 & 5 & 6 \end{bmatrix}$
Column vector	$n \times 1$	$\begin{bmatrix} 1 \\ 4 \end{bmatrix}$
Row vector	$1 \times p$	$[1 \quad 2 \quad 3]$
Scalar	1×1	$[1]$
Square	$n \times n$	$\begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$
Symmetric	Square $a_{ij} = a_{ji}$	$\begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$
Diagonal	Square $a_{ij} = 0$ for $i \neq j$	$\begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$
Identity (**I**)	Square $a_{ij} = 1$ $a_{ij} = 0$ for $i \neq j$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Upper/lower triangular	Square $a_{ij} = 0$ below/above the diagonal	$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & 9 \end{bmatrix}$
Unit vector (**1**)	$a_i = 1$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Unit matrix (**J**)	$a_{ij} = 1$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
Null(zero) matrix (**0**)	$a_{ij} = 0$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Operations of Matrices

➢ **Addition and Subtraction of Matrices**

Matrices can be added and subtracted if (and only if) they possess the same dimensions.

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} + \begin{bmatrix} j & m & p \\ k & n & q \\ l & o & r \end{bmatrix}$$

$$\begin{bmatrix} a+j & d+m & g+p \\ b+k & e+n & h+q \\ c+l & f+o & i+r \end{bmatrix}$$

Matrix Operations

- **Matrix Equality**
- Let A and B be two matrices.
  $A=B$ if they have the **same number of rows and columns**, and every element at each position in A equals element at corresponding position in B.
- Matrices having equal corresponding entries.

- **Examples**

$$\begin{bmatrix} 5 & 0 \\ -\frac{4}{4} & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -1 & 0.75 \end{bmatrix} \qquad \begin{bmatrix} -2 & 6 \\ 0 & -3 \end{bmatrix} \neq \begin{bmatrix} -2 & 6 \\ 3 & -2 \end{bmatrix}$$

Operations of Matrices

Scaler x Multiplied to a Matrix

Scalar Multiplication

$$x\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ax & bx \\ cx & dx \end{bmatrix}$$

Matrix Multiplication

$$c_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} + a_{14}b_{41}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ b_{41} & b_{42} & b_{43} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix}$$

2 x 4 4 x 3 2 x 3

$$c_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} + a_{24}b_{42}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ b_{41} & b_{42} & b_{43} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix}$$

Important remarks

(Find example wherever possible)

- In general, $AB \neq BA$.
- If $AB = 0$ does not imply that $A = 0 \; or \; B = 0$.
- If $AB = AC$ does not imply that, $B = C$.
- Distributive Law holds for Matrix multiplication

  $$A(B + C) = AB + AC, \qquad (A + B)C = AC + BC$$
- $A^2 = A \times A, \; A^3 = A^2 \times A \;\; i.e \;\; (A^m)^n = A^{mn}$
- $A^m A^n = A^{m+n}$
- $(A')' = A$
- $(AB)' = B'A'$
- $(A + B)' = A' + B'$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $|AB| = |A|.|B|$

Rank of a Matrix

The matrix is said to be of rank r if there is

1. At least one minor of the order r which is not equal to zero
2. Every minor of order (r+1) is equal to Zero.

Rank is denoted by $\rho(A)$.

$Ex. A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ Note that |A|=0 and also any minor of order 2×2 is 0. Hence, largest order of its non vanishing minor is 1.

Remark: $\rho(A_{m\times n}) \leq \min(m, n)$

Ways to find Rank

Echelon Form

Given Matrix is to be converted to Row Echelon form using **only elementary row operations.** Then, number of non-zero rows represents rank of Matrix.

Normal Form

Given matrix is converted to normal form(roughly identity matrix) **using both elementary row and column operations**. Then the order of (roughly) identity matrix is rank of matrix.

ECHELON FORM

- A rectangular matrix is in **echelon form** (or **row echelon form**) if it has the following three properties:
  1. All nonzero rows are above any rows of all zeros.
  2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
  3. All entries in a column below a leading entry are zeros.

Normal Form

By performing elementary **row and column transformations**, any non-zero matrix A can be reduced to one of the following four forms, called the **Normal form**.

- $[I_r]$
- $[I_r \;\; 0]$
- $\begin{bmatrix} I_r \\ 0 \end{bmatrix}$
- $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

The number ‘r‘so obtained is called the rank of the Matrix.

System of Linear Equations

- **A system of *m* linear equations in *n* variables** is a set of *m* equations, each of which is linear in the same *n* variables:

$$\begin{aligned}
a_{11}x_1 &+ a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\
a_{21}x_1 &+ a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\
a_{31}x_1 &+ a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_3 = b_3 \\
&\vdots \\
a_{m1}x_1 &+ a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_m
\end{aligned}$$

Number of Solutions of a System of Linear Equations

For a system of linear equations in *n* variables, precisely one of the following is true.

1. The system has exactly one solution (consistent system).
2. The system has an infinite number of solutions (consistent system).
3. The system has no solution (inconsistent system).

Matrix form of System of Linear Equations

Matrix form: $Ax = b$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

In above form,

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & & & & \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

is coefficient Matrix.

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ vector of unknowns}$$

and

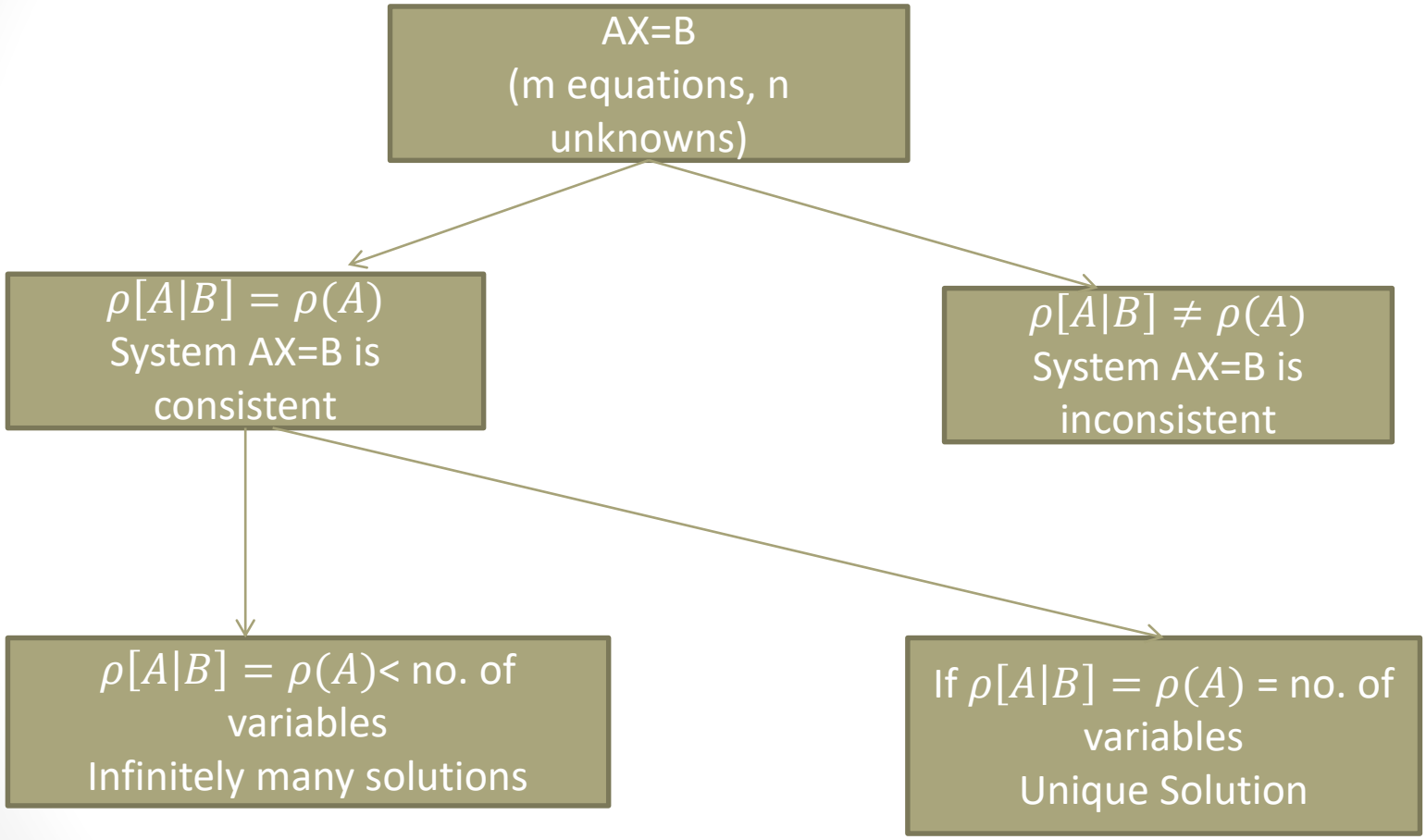
$$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}, \text{ right hand side}$$

then the system can be written as

$$AX = B$$

The matrix $\left[A \middle| B\right]$ or $\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right]$ is called the **augmented matrix**.

Solution of system of Linear Equations



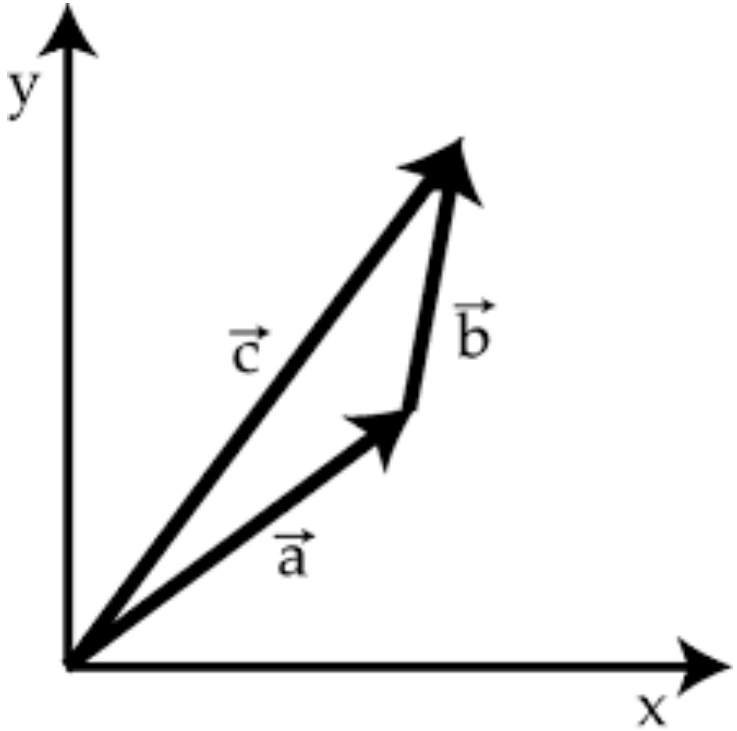
Solution of System AX=B(Method)

1. Write the given system in matrix form(AX=B).
2. Consider the augmented matrix[A|B] from the given system.
3. Reduce the augmented matrix to the Echelon form.

Note : In [A|B], first part represents $\rho(A)$ and whole matrix represents $\rho(A|B)$.

4. Conclude the system has unique, infinite or no solution.
5. If consistent with $\rho(A|B) = \rho(A) =$ no. of variables then rewrite equations and find values.
6. If consistent with $\rho(A|B) = \rho(A) = r <$ no. of variables then put $n - r$ variables(free variables) as u, v, w etc. find values of other variables in terms of free variables.

Linear Dependence and Independence



By Triangle law,

$$\vec{c} = \vec{a} + \vec{b}$$

Set $\{\vec{a}, \vec{b}, \vec{c}\}$ is Linearly dependent.

Set $\{\vec{a}, \vec{b}\}$ is linearly Independent.

Can you think of the definition of Linearly dependent and Linearly Independent?

Definition

Linearly Independent

The system n vectors Set $x_1, x_2, x_3 \ldots \ldots . x_n$ is said to be Linearly Independent if every relation of the type

$c_1 x_1 + c_2 x_2 + \cdots .. c_n x_n = 0$

has a unique solution $c_1 = c_2 = \ldots \ldots . = c_n = 0$

Linearly Dependent

The system n vectors Set $x_1, x_2, x_3 \ldots \ldots . x_n$ is said to be Linearly dependent if every relation of the type

$c_1 x_1 + c_2 x_2 + \cdots .. c_n x_n = 0$

Has some $c_i \neq 0$.

Remarks

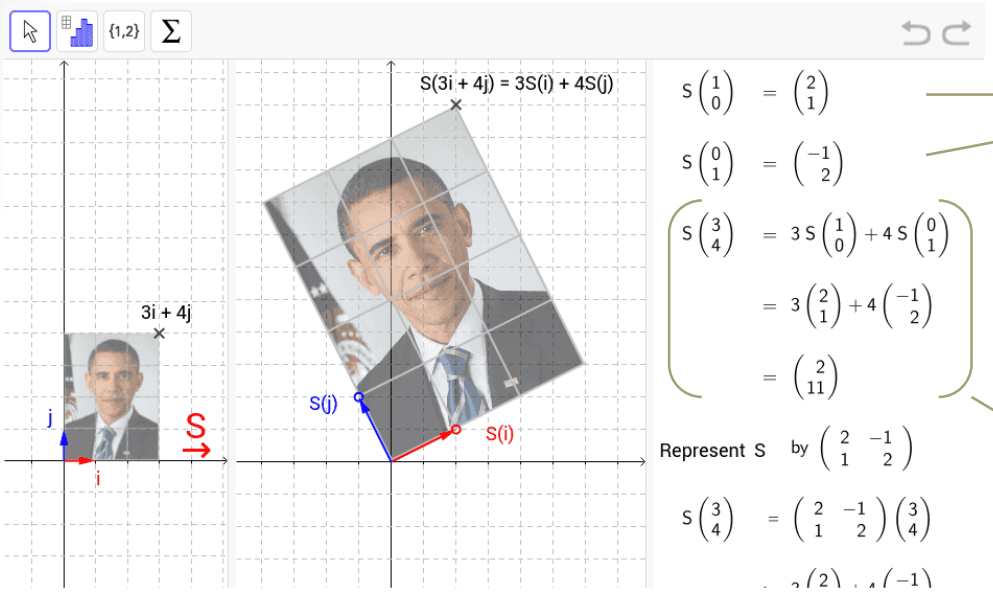
- $c_1x_1 + c_2x_2 + \cdots .. c_nx_n = 0$ is a linear equation with c_1, c_2, c_3 … … c_n as unknowns.
- If vectors x_1, x_2, x_3, … … x_n are linearly dependent then in equation $c_1x_1 + c_2x_2 + \cdots .. c_nx_n = 0$ by definition some $c_k \neq 0$.
- By simplification we get

$$c_kx_k = -c_1x_1 - c_2x_2 \ldots\ldots - c_nx_n$$

$$x_k = -\left(\frac{c_1}{c_k}\right)x_1 - \left(\frac{c_2}{c_k}\right)x_2 - \ \ldots\ldots\ldots\ldots - \left(\frac{c_n}{c_k}\right)x_n$$

- ***If vectors are dependent then there exist a relation between the given vectors.***

Linear Transformation



Matrix S is applied to vectors $\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}$

Matrix S transforms each and every vector on the plane

A Linear Transformation is a map from 'n' dimensional space to itself generally represented by

$$y_1 = a_{11}x_1 + a_{12}x_2 \ldots . + a_{1n}x_n$$
$$y_2 = a_{21}x_1 + a_{22}x_2 \ldots . + a_{2n}x_n$$
$$.$$
$$.$$
$$y_n = a_{n1}x_1 + a_{n2}x_2 \ldots . + a_{nn}x_n$$

Matrix Representation of Linear Transformation

- Matrix representation of the above Linear Transformation is given by

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

- Every Linear Transformation is the form Y = AX (**Expressing Y in terms of X**)
- Properties of Linear Transformation depends on matrix 'A'
- If $|A| = 0$, then matrix A is called singular and transformation is called Singular transformation.
- If $|A| \neq 0$, then matrix A is called non-singular and transformation is called non-singular or regular transformation.

Orthogonal Transformation

- Transformation $Y = AX$ is said to be Orthogonal if A is Orthogonal matrix.
- Matrix A is called orthogonal if $A'A = AA' = I$ where A' is transpose of A,.

Properties of Orthogonal Matrix :

1. $A^{-1} = A'$
2. $|A| = \pm 1$
3. A is Orthogonal then A^{-1} is also Orthogonal.

Eigen Values and Eigen Vectors

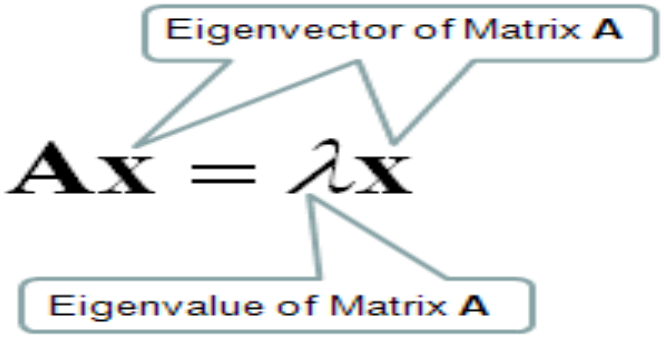


- Observe blue and red vectors before and after the transformation.

Blue vector has same direction even after the transformation where as red vector changes its direction.

Eigen Value and Eigen Vectors

- **Definition:** An **eigenvector** of an $n \times n$ matrix A is a nonzero vector **x** such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution **x** of $A\mathbf{x} = \lambda\mathbf{x}$; such an **x** is called an *eigenvector corresponding to* λ.



Remember
Blue vector?

Graphically Eigen vector is the one that doesn't change its direction
And Eigen values are the extent by which Eigen vector changes its length.

Let x be an eigenvector of the matrix **A.** Then there must exist an eigenvalue λ such that $Ax = \lambda x$ or, equivalently,

$$Ax - \lambda x = 0 \quad \text{or}$$

$$(A - \lambda I)x = 0$$

If we define a new matrix $B = A - \lambda I$, then

$$Bx = 0$$

If **B** has an inverse then $x = B^{-1}0 = 0$. But an eigenvector cannot be zero.

Thus, it follows that x will be an eigenvector of **A** if and only if **B** does not have an inverse, or equivalently **det(B)=0**, or

$$\det(A - \lambda I) = 0$$

This is called the **characteristic equation** of **A**. Its roots determine the eigenvalues of **A**.

Characteristic Equation

- Characteristic Equation is also called characteristic Polynomial

because $|A - \lambda I| = 0$ is a polynomial of degree equal to order of square matrix A.

- For 2×2 matrix, characteristic polynomial is given by

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\lambda^2 - S_1\lambda + |A| = 0$$

Where S_1 = sum of diagonal matrix = Trace (A)

- For 3×3 matrix, characteristic polynomial is given by

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

Where S_1 = sum of diagonal matrix =Trace(A)

S_2 = sum of minors of order two of diagonal elements

$$S_2 = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

Method

(To Eigen values and vectors for a given matrix 'A')

1. Write the characteristic polynomial for 'A' i.e $|A - \lambda I| = 0$. (use the shortcuts for 2×2 and 3×3 matrix)
2. Roots of the Characteristic polynomial are Eigen Values. (say λ_1, $\lambda_2, \lambda_3 \ldots\ldots$)
3. For each Eigen Value λ_i, we can find Eigen vectors by solving the system of linear equation

$$[A - \lambda_i I]X = 0$$

Brain Teaser

Can same Eigen vector can correspond to two different Eigen Values?
If no, then why? If yes, can you find it?
HINT: USE THE BASIC DEFINITION OF EIGEN VECTOR

Eigenvalue Properties

- Eigenvalues of $\mathbf{A}$ and $\mathbf{A}^T$ are equal
- Singular matrix has at least one zero eigenvalue
- Eigenvalues of $\mathbf{A}^{-1}$: $1/\lambda_1, 1/\lambda_2, \ldots, 1/\lambda_n$
- Eigenvalues of diagonal and triangular matrices are equal to the diagonal elements
- Trace

$$Tr(\mathbf{A}) = \sum_{j=1}^{n} \lambda_j$$

- Determinant

$$|\mathbf{A}| = \prod_{j=1}^{n} \lambda_j$$

Cayley-Hamilton Theorem

Statement: Every square matrix satisfies its own characteristic equation.

We know that characteristic equation of a square matrix is

$$|A - \lambda I| = 0$$

i.e. $a_0\lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n = 0$

C-H theorem says that A satisfies above equation, which means if we replace λ by matrix A, we will get null matrix on RHS.

i.e. $a_0A^n + a_1A^{n-1} + \cdots + a_{n-1}A + a_nI = [0]$

Where I is identity matrix of size n

- **Applications of Cayley-Hamilton Theorem:**

1. Nth power of any square matrix can be expressed as linear combination of lower powers of A.
2. To find inverse of non-singular matrix A.

Ex. 1 : *Verify Cayley-Hamilton theorem for* $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ *and use it to find* A^4 *and* A^{-1}.

Sol. : **Step 1 :** The characteristic equation of A is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 2 & -2 \\ -1 & 3-\lambda & 0 \\ 0 & -2 & 1-\lambda \end{vmatrix} = 0.$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

$$S_1 = 1 + 3 + 1 = 5$$

$$S_2 = \begin{vmatrix} 3 & 0 \\ -2 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -1 & 3 \end{vmatrix}$$

$$= 3 + 1 + 5 = 9$$

$$|A| = 1(3) + 1(-2) = 1$$

Hence the characteristic equation of A is $\lambda^3 - 5\lambda^2 + 9\lambda - 1 = 0$.

rotation and translation to be applied to the model coordinate definition of the picture. The windowing transformation is a particular case in which no rotation is applied.

Windowing transformation consists of scaling and translation.

Scaling : If (x, y, z) are the coordinates of a point P with reference to ox, oy, oz as coordinate axes (see Fig. 3.1) and (X, Y, Z) are the coordinates of a point P with reference to OX, OY, OZ as coordinate axes and if t is the scaling factor, then,

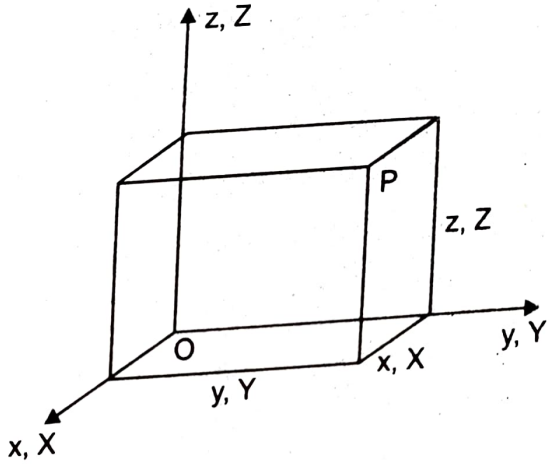


Fig. 3.1

$$X = tx \quad \ldots (1)$$

$$Y = ty$$

$$Z = tz$$

or in matrix form

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Translation : If (x, y, z) are the coordinates of the point P w.r.t. ox, oy, oz and (X, Y, Z) are the coordinates of the point P w.r.t. O'X, O'Y, O'Z (See Fig. 3.2), where O' has coordinates (dx, dy, dz) w.r.t. ox, oy, oz then,

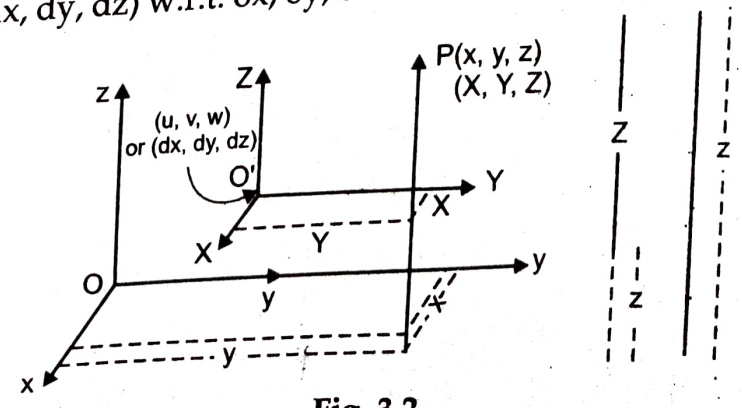


Fig. 3.2

$$X = x - dx \qquad X = x - u$$
$$Y = y - dy \quad \text{OR} \quad Y = y - v \qquad \ldots(2)$$
$$Z = z - dz \qquad Z = z - w$$

To express these relations in matrix transformation form, we use homogeneous coordinates.

This involves representing the position of a point w.r.t. ox, oy, oz by (ux, uy, uz, u) instead of (x, y, z) and instead of representing the point by (X, Y, Z), we represent it as (uX, uY, uZ, u) where the scale factor u is normally taken as unity. Now relations (2) can be written in matrix form as

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -dx \\ 0 & 1 & 0 & -dy \\ 0 & 0 & 1 & -dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

OR

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -u \\ 0 & 1 & 0 & -v \\ 0 & 0 & 1 & w \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

(1) If we combine the rotation about z-axis and translation then we get the new co-ordinates as [i.e. if the rotation is about z-axis by angle θ and origin is shifted to (u, v, w) then]

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & -u \\ -\sin\theta & \cos\theta & 0 & -v \\ 0 & 0 & 1 & -w \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

(2) If rotation is about x-axis and origin is shifted to (u v w) then

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -u \\ 0 & \cos\theta & \sin\theta & -v \\ 0 & -\sin\theta & \cos\theta & -w \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

(3) If rotation is about y-axis and origin is shifted to (u, v, w) then

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta & -u \\ 0 & 1 & 0 & -v \\ +\sin\theta & 0 & \cos\theta & -w \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Rotation :

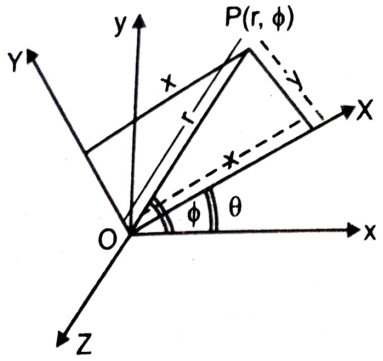


Fig. 3.3

Consider a point P whose coordinates w.r.t. ox, oy, oz are (x, y, z) and w.r.t. OX, OY, OZ are (X, Y, Z). Consider rotation taking place about the z-axis of the model coordinate system as shown in Fig. 3.3. It is clear that z coordinates in both systems will remain same.

Let (r, ϕ) be polar coordinates of the point P in xoy plane.

$$\therefore \quad x = r\cos\phi, \quad y = r\sin\phi \qquad \ldots (4)$$

If angle of rotation of the axes is θ, then

$$X = r\cos(\phi - \theta), \quad Y = r\sin(\phi - \theta)$$

$$\therefore \quad X = r[\cos\phi\cos\theta + \sin\phi\sin\theta] \qquad \ldots (5)$$

$$Y = r[\sin\phi\cos\theta - \cos\phi\sin\theta]$$

$$\therefore \quad X = (r\cos\phi)\cos\theta + (r\sin\phi)\sin\theta \qquad \ldots (6)$$

$$Y = (r\sin\phi)\cos\theta - (r\cos\phi)\sin\theta$$

$$\therefore \quad X = x\cos\theta + y\sin\theta \qquad \ldots (7)$$

$$Y = -x\sin\theta + y\cos\theta$$

$$Z = z \quad \text{[As rotation is about z-axis]}$$

Relation (7) can be put in matrix form as

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad \ldots (8)$$

Similarly, for rotation about x-axis, we shall obtain

$$\ldots (9)$$

$$X = x$$

$$Y = y\cos\theta + z\sin\theta$$

$$Z = -y\sin\theta + z\cos\theta$$

or, in matrix form,

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \ldots (10)$$

And rotation about y-axis will give

$$\begin{aligned} Y &= y \\ X &= -z\sin\theta + x\cos\theta \\ Z &= z\cos\theta + x\sin\theta \end{aligned} \quad \ldots (11)$$

or, in matrix form,

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \ldots (12)$$

Ex. 1 : *Co-ordinates of a point P are (50, 50, 50). Origin is shifted to the point (5, –2, 3). Rotation is about z-axis through 45°. Find the co-ordinates of P in new co-ordinate system.*

Sol. : As the rotation is about z-axis and translation is **(u, v, w), thus the rotation and** translation together in matrix form is given by

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 & -u \\ -\sin\theta & \cos\theta & 0 & -v \\ 0 & 0 & 1 & -w \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

Substituting $x = 50, y = 50, z = 50, u = 5, v = -2, w = 3, \theta = 45°$, we get

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 & -5 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & 2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 50 \\ 50 \\ 50 \\ 1 \end{bmatrix}$$

Thus,

$$X = \frac{50}{\sqrt{2}} + \frac{50}{\sqrt{2}} + 0 - 5$$

$$Y = \frac{-50}{\sqrt{2}} + \frac{50}{\sqrt{2}} + 0 + 2$$

$$Z = 0 + 0 + 50 - 3$$

$$1 = 1$$

$$\therefore \quad X = 65.71, \; Y = 2, \; Z = 47.$$

Ex. 2 : *Centre of the arc of the circle is (10, 10, 0) in a given coordinate system. Origin is shifted to the point (2, 0, 0) along with coordinate axes and then rotation is carried about x-axis through 60°. Find the coordinates of the centre in new coordinate system.*

Sol. : Let (x, y, z) be the coordinates of the centre in old system and X, Y, Z be the coordinates of the centre in new system.

Using relations (10), (3) and homogeneous coordinates, we get, considering only rotation

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 60^\circ & \sin 60^\circ & 0 \\ 0 & -\sin 60^\circ & \cos 60^\circ & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \ldots (1)$$

Considering only translation

or
$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \ldots (2)$$

Combining transformations (1) and (2),

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 0.50 & 0.866 & 0 \\ 0 & -0.866 & 0.5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad \ldots (3)$$

Now putting $x = 10$, $y = 10$, $z = 0$ in (3) and simplifying, we get,

$$X = 10 - 2 = 8$$

$$Y = 5$$

$$Z = -8.66$$

$\therefore$ Coordinates of centre in new system are (8, 5, – 8.66).

Ex. 3 : *Centre of arc of the circle is (10, 10, 10). Origin is (0, 0, 0), rotation is about through an angle 60°. Find the centre of arc of the circle in new co-ordinate system.*

Sol. : As the rotation is about x-axis, thus

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Given : $x = 10, y = 10, z = 10$ and $\theta = 60°$

As the origin is (0, 0, 0),

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 60 & \sin 60 \\ 0 & -\sin 60 & \cos 60 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & \sqrt{3}/2 \\ 0 & -\sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ 10 \end{bmatrix}$$

$X = 10, Y = 13.66, Z = -3.66$

EXERCISE 3.8

1. Coordinates of a point P are (50, 50, 50). Origin is shifted to the point (5, –2, 3) with axes transferred parallel to themselves. Rotation is carried out about z-axis through 45°. Find the coordinates of the point P in new coordinate system. **[Ans.** (65.7, 2, 47)]
2. Centre of the arc of the circle in a given coordinate system is (100, 100, 100). Origin is shifted to the point (–10, –5, 2). Rotation is carried out about y-axis through an angle of 30°. Find the centre of the arc of the circle in new coordinate system. **[Ans.** (46.66, 105, 134.66)]

   Note : The use of homogeneous coordinates has many applications in computer graphics, in addition to the generations of viewing transformations, already discussed. Translational and rotational transformations are also very useful in robotics and in mechanisms analysis. Discussion of these usages is beyond the scope of this work.
3. *Centre of the arc of the circle is (10, 10, 10) in a given co-ordinate system. Origin (0 0 0) and then rotation is carried about x-axis through 60°. Find the co-ordinates of centre in the new co-ordinate system.* **[Ans.** (10, 5, – 8.66)]
4. *Centre of the arc of the circle is (10, 10, 10). Origin is shifted to the point (2, 0, 0) along with co-ordinate axes and then rotation is carried out about x-axis through 60°. Find co-ordinates of the centre in new co-ordinate system.* **[Ans.** (8, 5, – 8.66)]
5. *Centre of arc of the circle in a given co-ordinate system is (100, 100, 100). Origin is shifted to the point (–10, –5, 2). Rotation is carried out about y-axis through an angle of 30°. Find the centre in new co-ordinate system.* **[Ans.** (46.66, 105, 134.76)]